IOT SDN NFV

Computer Vision

6_9_3_Netwotk Architecture and OSI

https://github.com/siagianp
https://www.youtube.com/@AmelOline/videos
https://github.com/amelcharolinesgn2/IoT_simulator-mqtt-NodeRed
https://github.com/amelcharolinesgn2/Cloud-Incomastructures
https://github.com/amelcharolinesgn2/ClouD-Incomastructure-SISKA-2025

Network Architecture

- **Network Architecture**

Networks **do not remain fixed** at single point in time, but it must evolve to accommodate changes based on the **technologies** on which they are based and demands made by **application programmer.**

- **Network architecture guides the design and implementation of network.** Two commonly used architecture are

OSI Architecture

Internet or TCP/IP architecture

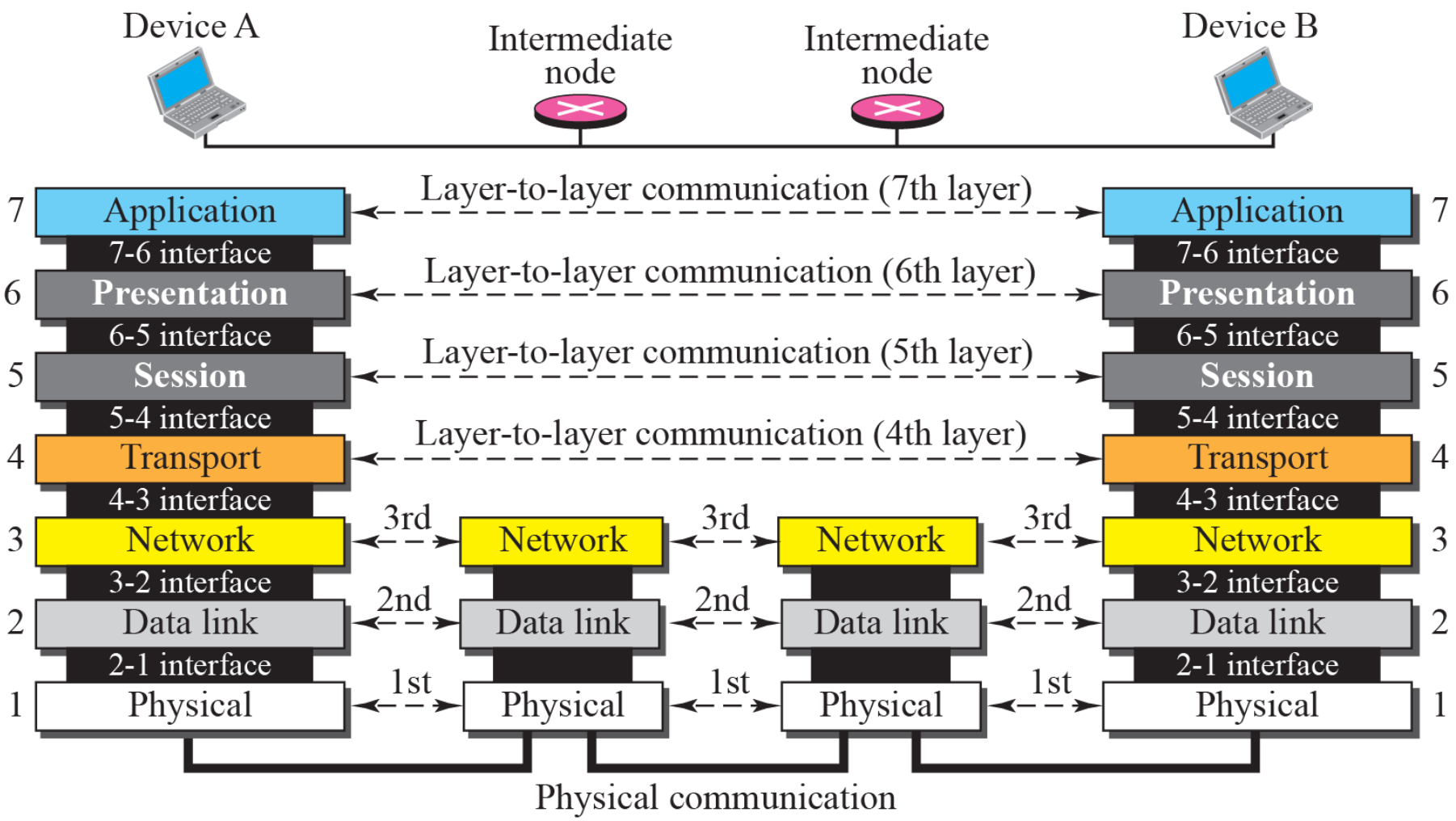
Features of Layer Architecture

- **Layer architecture simplifies the network design.**
- **It is easy to debug network applications in a layered architecture network.**
- **The network management is easier due to the layered architecture.**
- **Network layers follow a set of rules, called protocol.**
- **The protocol defines the format of the data being exchanged, and the control and timing for the handshake between layers.**
- **A protocol is a set of rules that governs data communication. It defines what is communicated, how it is communicated, and when it is communicated**
- It provides more **modular design**. To add some new service, it is enough to modify the functionality at one layer, reusing the functions provided at all the other layers

Open Systems Interconnection (OSI) Model

- International standard organization (ISO) established a committee in 1977 to develop an architecture for computer communication.
- Open Systems Interconnection (OSI) reference model is the result of this effort.
- In 1984, the Open Systems Interconnection (OSI) reference model was approved as an international standard for communications architecture.
- Term "open" denotes the ability to connect any two systems which conform to the reference model and associated standards.

Interaction between layers in the OSI model



Description of Layers

- **Network Support Layers**
  - Layers 1,2,3 - Physical, Data link and Network are the network support layers.
  - They deal with the **physical aspects of moving data from one device to another such as electrical specifications, physical addressing, transport timing and reliability**.
- **Transport Layer**
  - Layer4, transport layer, ensures **end-to-end reliable data transmission on a single link**.
- **User Support Layers**
  - Layers 5,6,7 – Session, presentation and application are the user support layers. They allow **interoperability among unrelated software systems**

Interaction between layers in the OSI model/Description of layers

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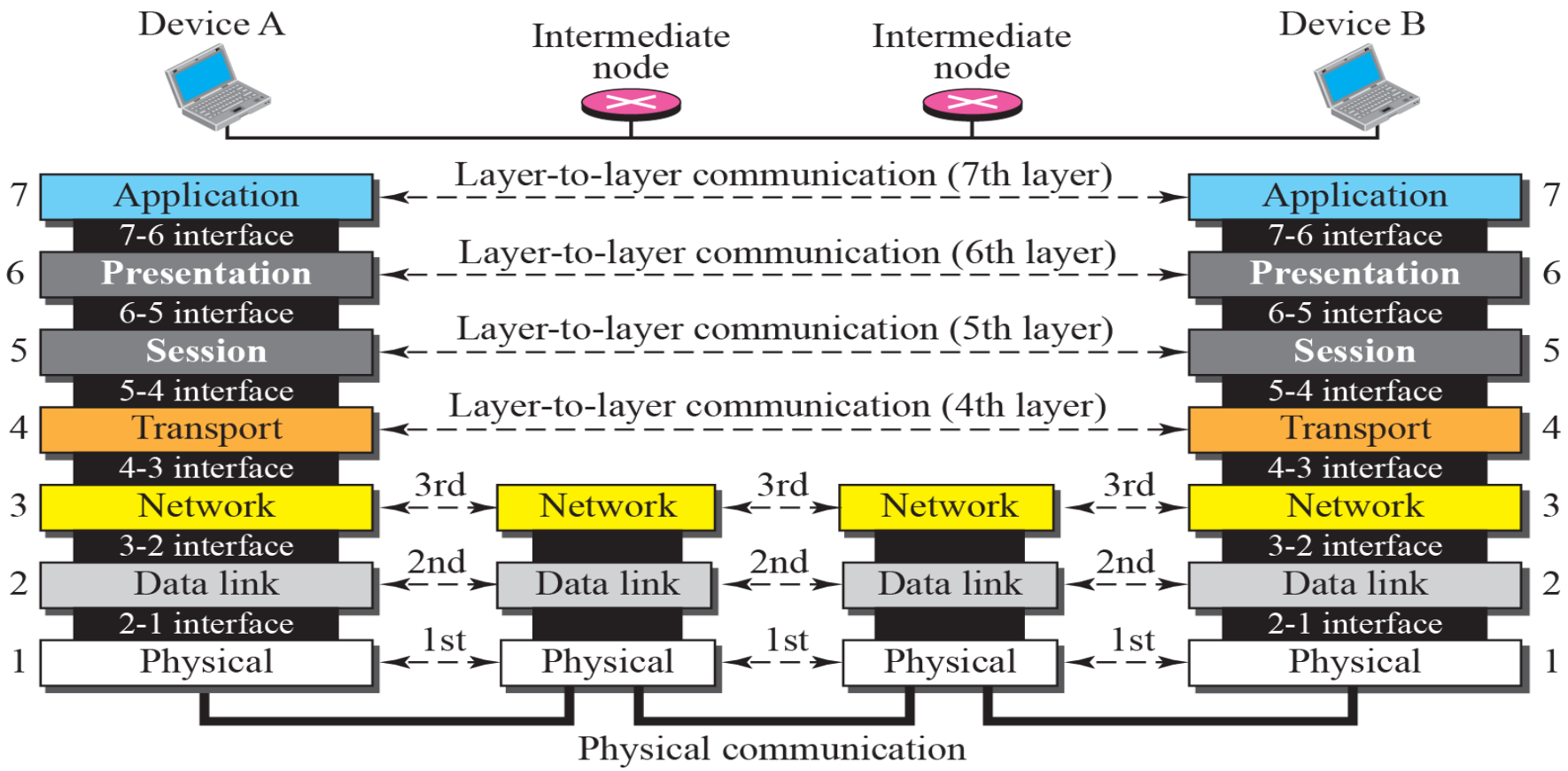
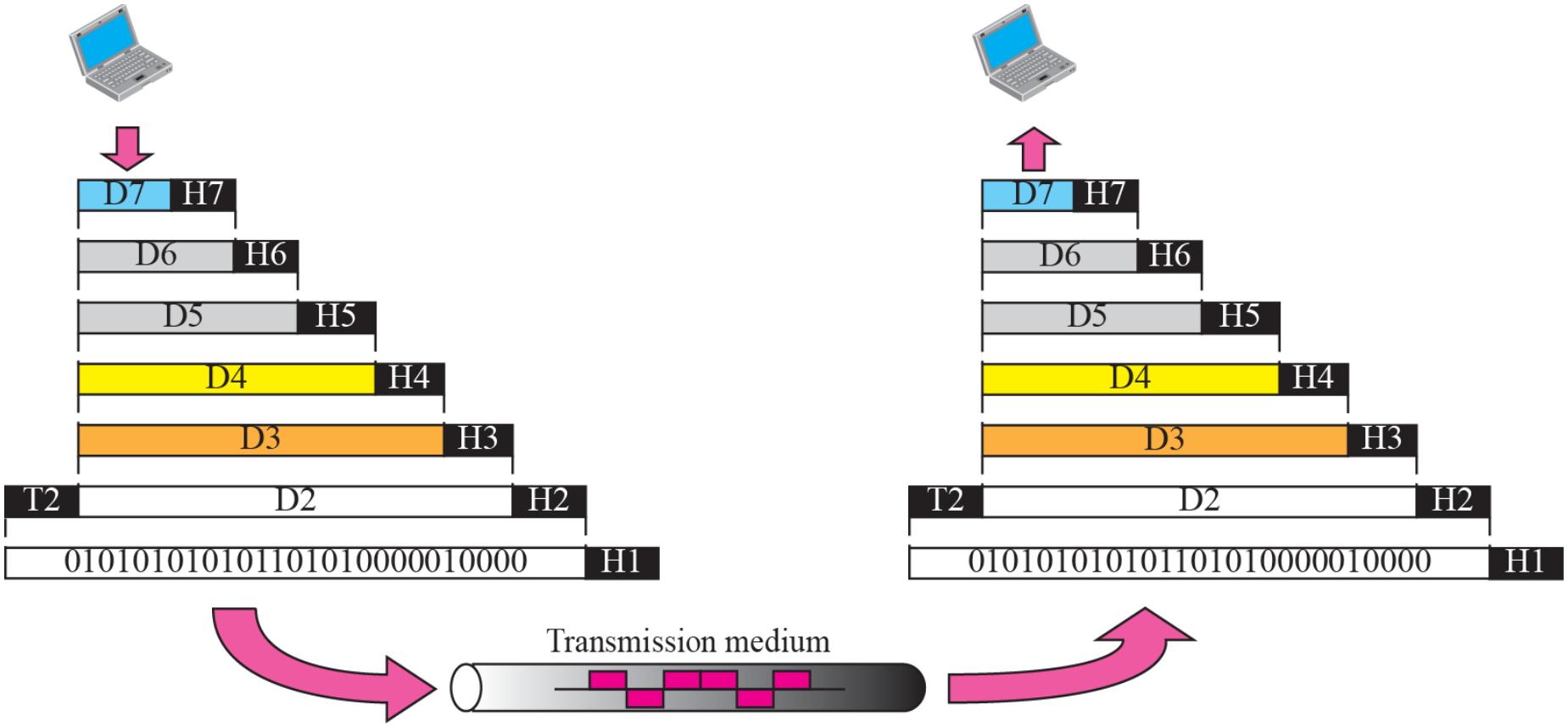


Figure *An exchange using the OSI model*



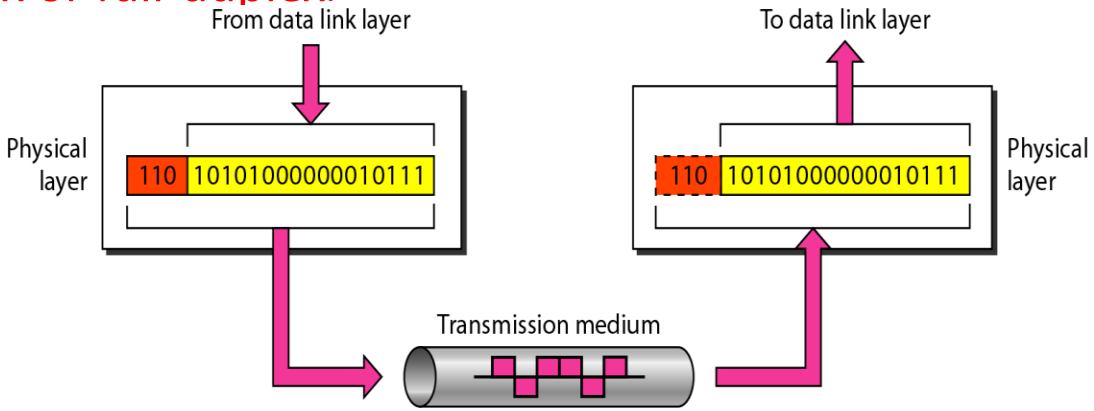
Summary of OSI Layers

Application	To allow access to network resources	7
Presentation	To translate, encrypt, and compress data	6
Session	To establish, manage, and terminate sessions	5
Transport	To provide reliable process-to-process message delivery and error recovery	4
Network	To move packets from source to destination; to provide internetworking	3
Data link	To organize bits into frames; to provide hop-to-hop delivery	2
Physical	To transmit bits over a medium; to provide mechanical and electrical specifications	1

Alligator
Pet
Steve's
Touch
Not
Do
Please

Physical layer

- **Physical characteristics of interfaces and media -** The physical layer defines the characteristics of the interface between the devices and the transmission medium.
- **Representation of bits -** To transmit the stream of bits, it must be encoded to signals. The physical layer defines the type of encoding.
- **Data Rate or Transmission rate -** The number of bits sent each second – is also defined by the physical layer.
- **Synchronization of bits -** The sender and receiver must be synchronized at the bit level. Their clocks must be synchronized.
- **Line Configuration -** In a point-to-point configuration, two devices are connected together through a dedicated link. In a multipoint configuration, a link is shared between several devices.
- **Physical Topology -** The physical topology defines how devices are connected to make a network. Devices can be connected using a mesh, bus, star or ring topology.
- **Transmission Mode -** The physical layer also defines the direction of transmission between two devices: simplex, half-duplex or full-duplex.

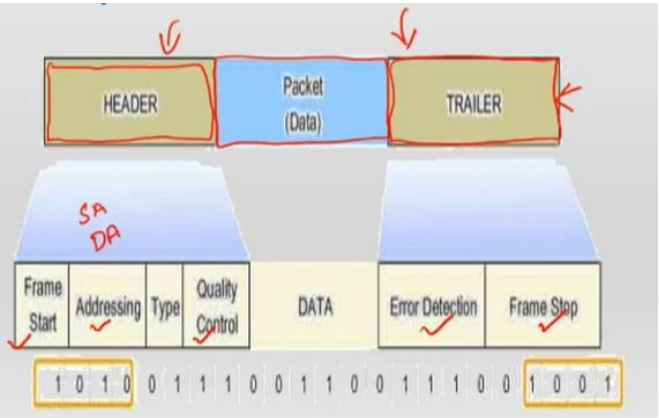


Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Data Link layer

- **It perform node to node packet delivery**
- **Framing -** Divides the stream of bits received into data units called frames.
- **Physical addressing** – If frames are to be distributed to different systems on the n/w , data link layer adds the source address and destination address in the header to the frame to identify the sender and receiver.
- **Flow control**- If the rate at which the data are absorbed by the receiver is less than the rate produced in the sender ,the Data link layer imposes a flow ctrl mechanism.
- **Error control**- Used for detecting and retransmitting damaged or lost frames and to prevent duplication of frames. This is achieved through a trailer added at the end of the frame.
- **Access control** -Used to determine which device has control over the link at any given time.



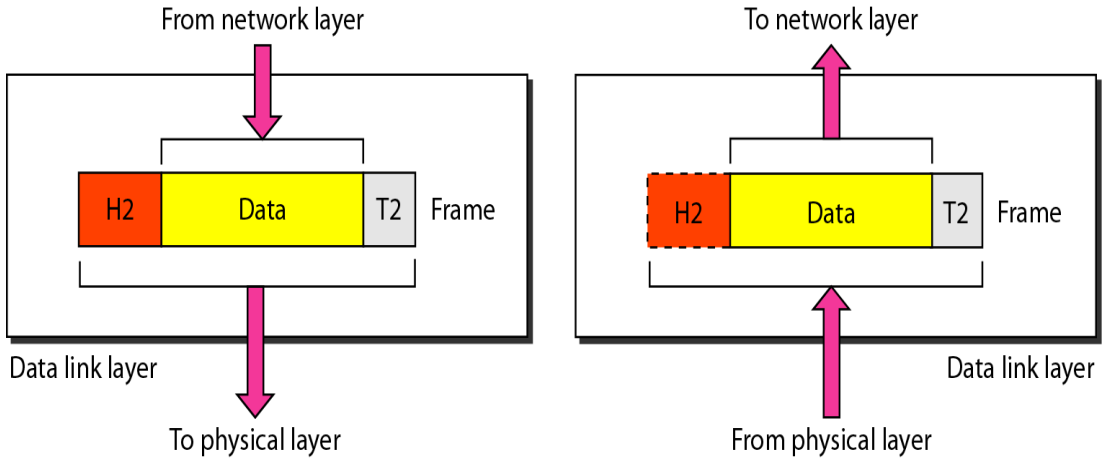
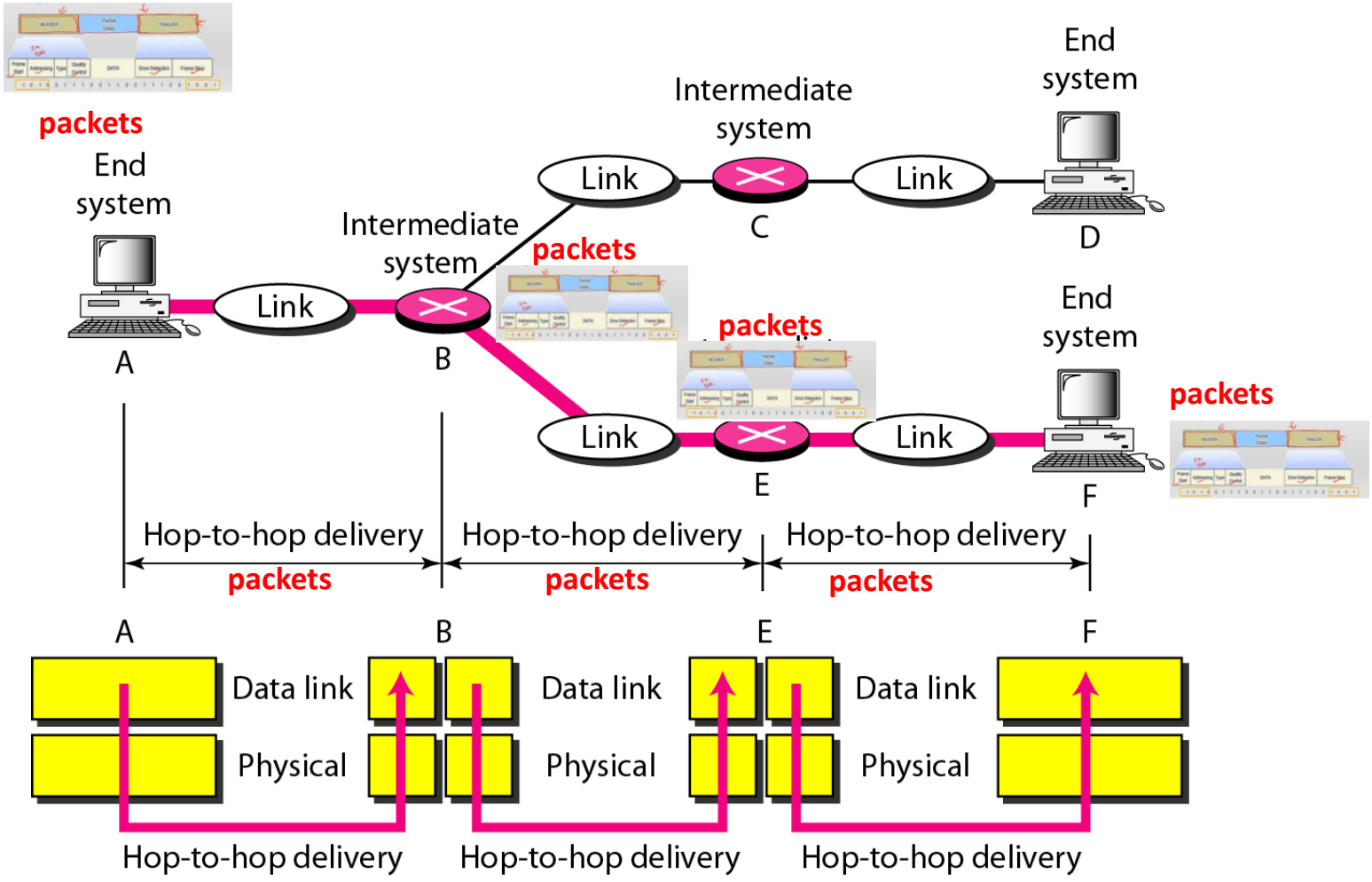


Figure *node-to-node delivery*

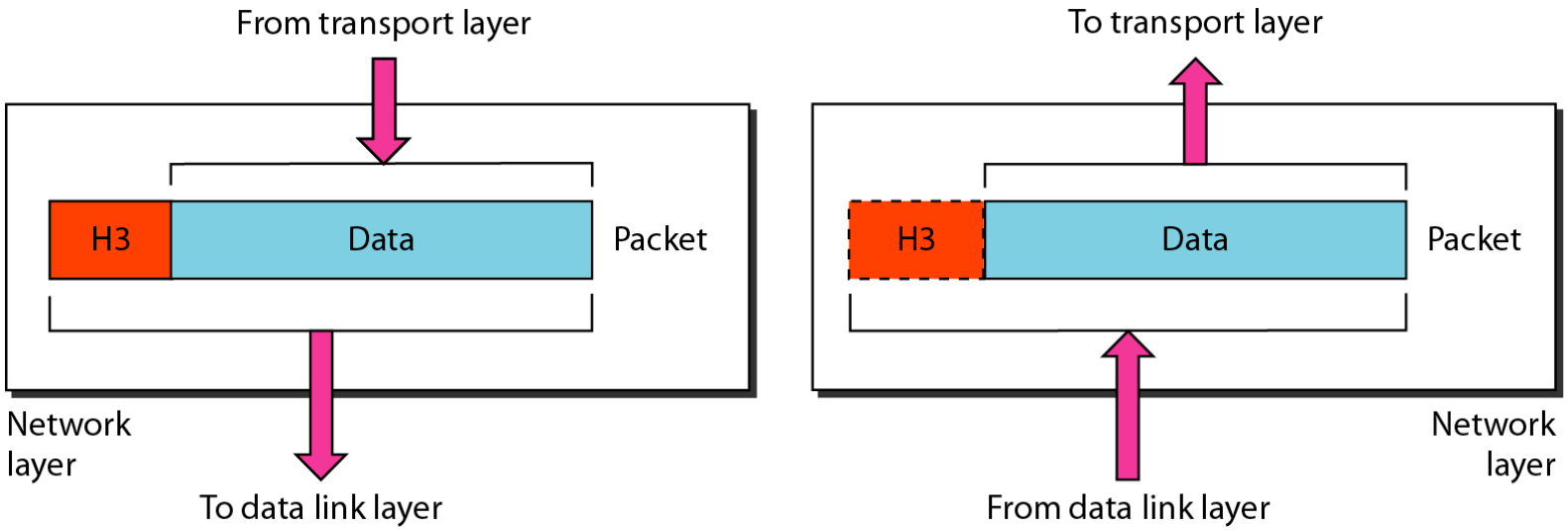


Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Network Layer

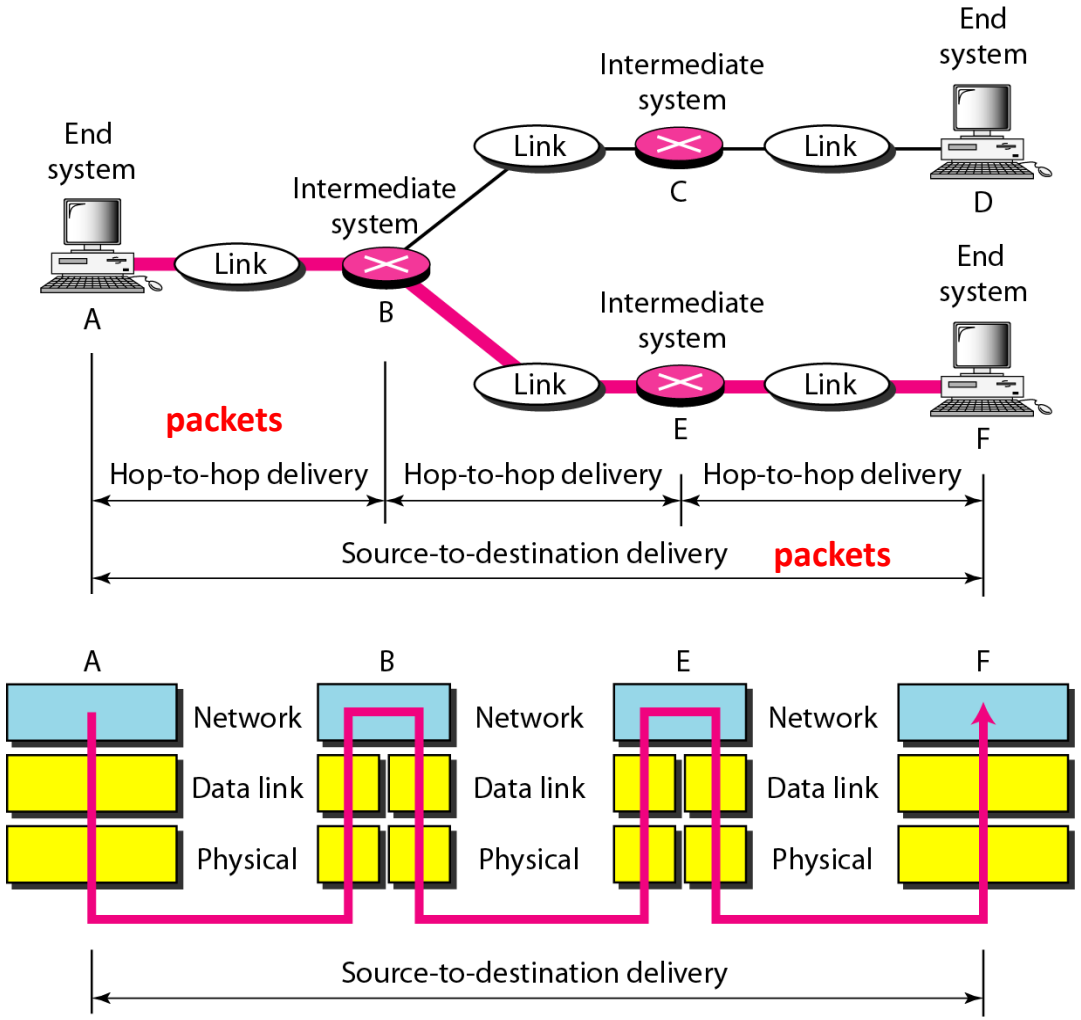
- This layer is responsible for the delivery of **packets from source to destination**
- **Logical addressing** - If a packet passes the n/w boundary, we need another addressing system for source and destination called logical address.
- **Routing** – The devices which connects various networks called routers are responsible for delivering packets to final destination.



Note

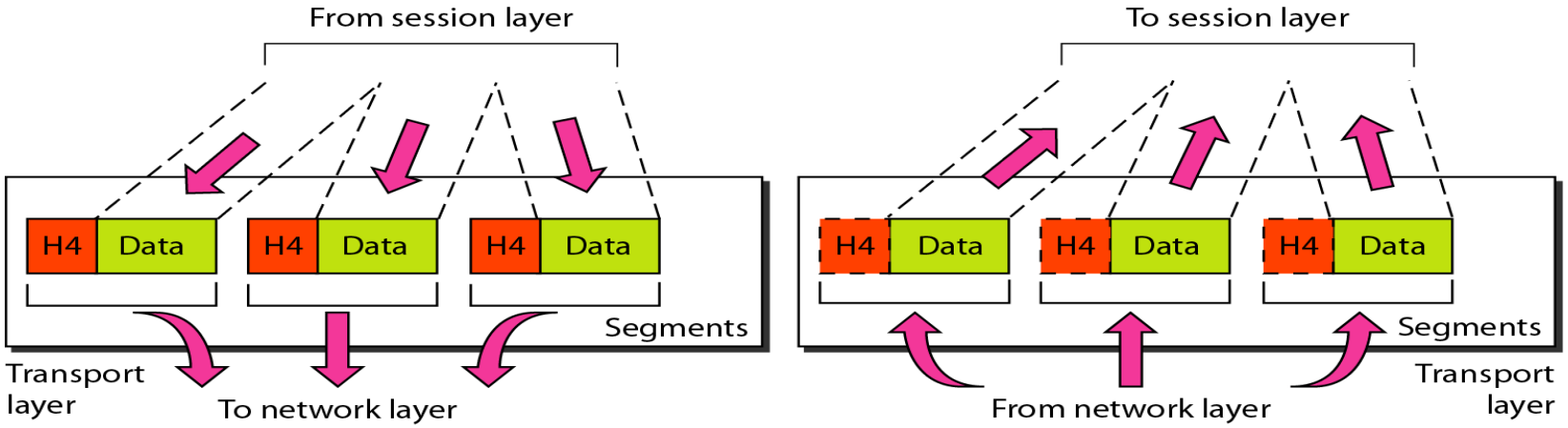
The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Figure *Source-to-destination delivery*

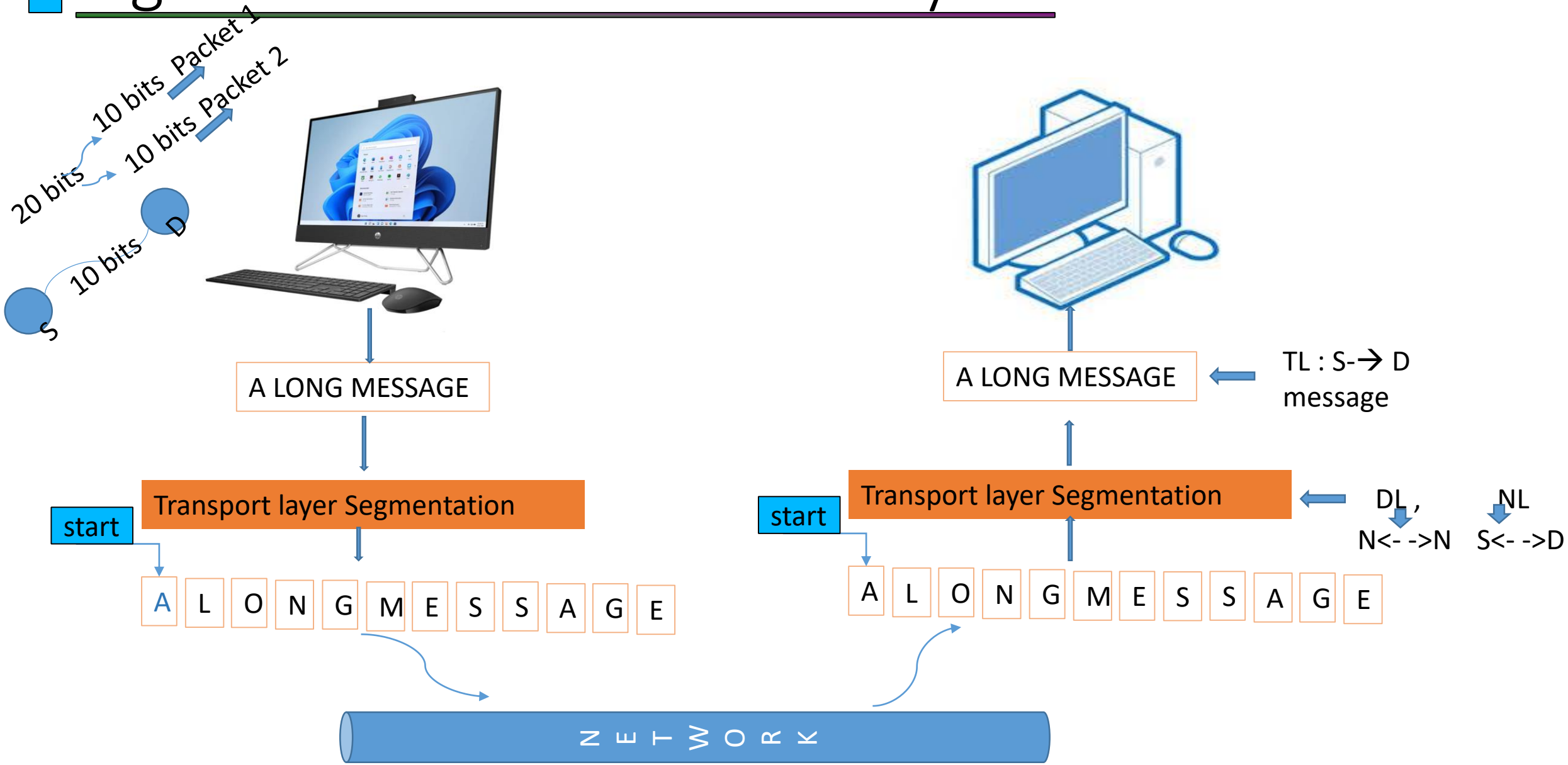


Transport Layer

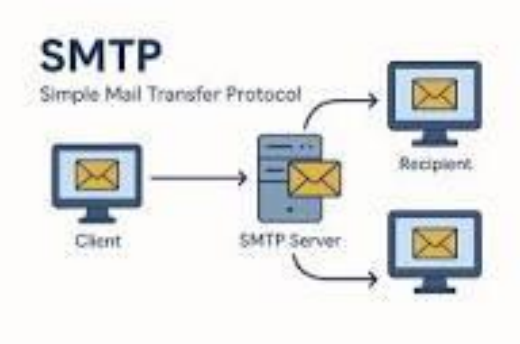
- **End to End message delivery** .
- **Segmentation and reassembly** - The message is divided into segments and each segment is assigned a sequence number. These numbers are arranged correctly on the arrival side by this layer.
- **Port addressing** - The header in this must therefore include a address called port address. This layer gets the entire message to the correct process on that computer.
- **Connection control** - This can either be **connectionless or connection-oriented.** The connectionless treats each segment as a individual packet and delivers to the destination. The connection-oriented makes connection on the destination side before the delivery. After the delivery the termination will be terminated.
- **Flow and error control** - Similar to data link layer, but process to process take place.



Segmentation and reassembly



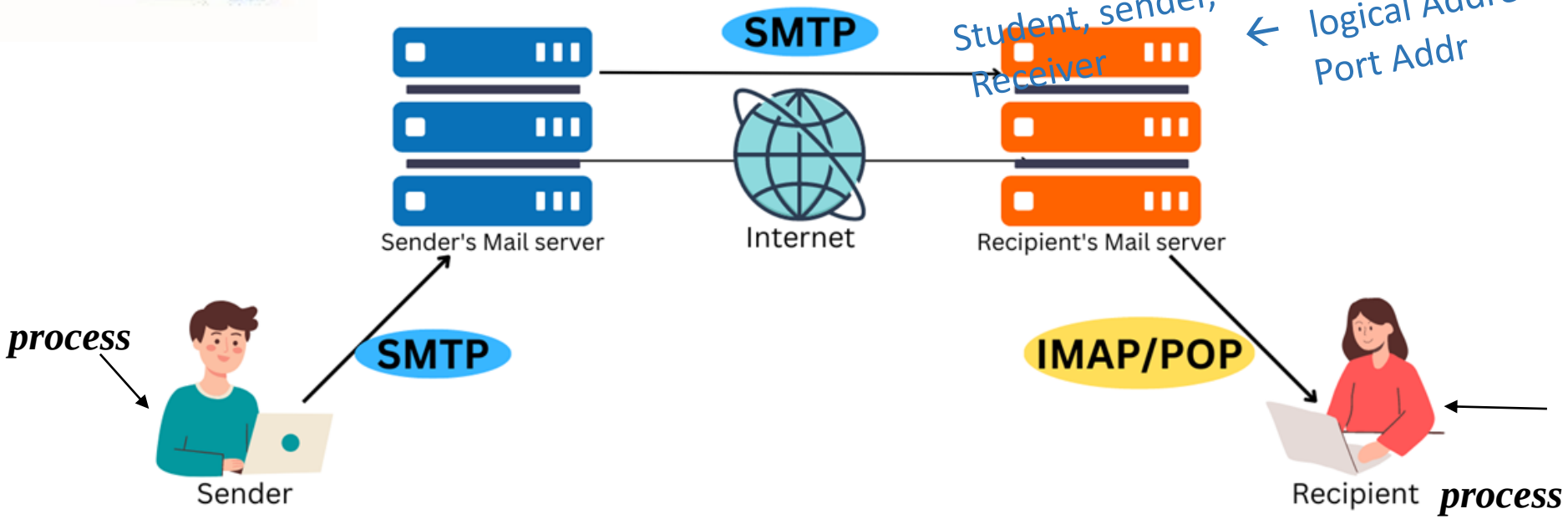
Reliable process-to-process delivery of a message



Service Port addressinge

Jame's ← Name ← Physical address → MAC – [NIC]

Student, sender, Receiver ← logical Address → Process ID
Port Addr



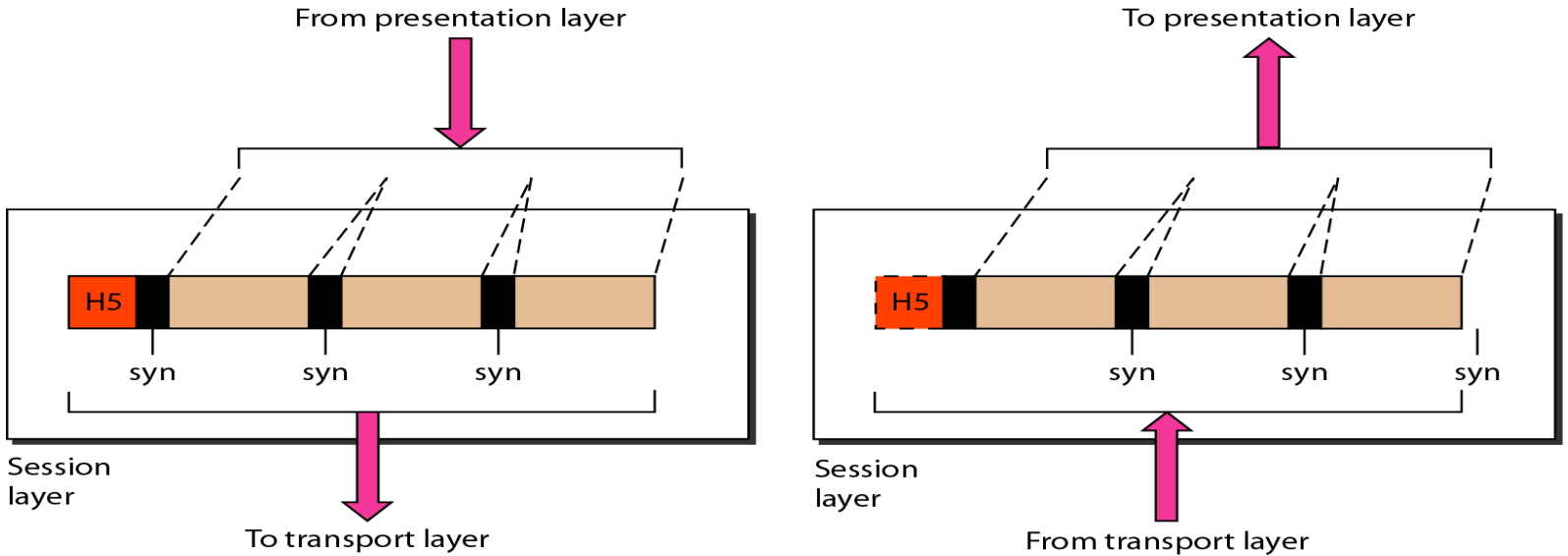
https://www.geeksforgeeks.org/computer-networks/network-address-translation-nat/

Note

The transport layer is responsible for the delivery of a message from one process to another.

Sessions Layer

- This layer establishes, session, manages and terminates connections between applications.
- **Dialog control** - This session allows two systems to enter into a dialog either in half duplex or full duplex.
- **Synchronization**-This allows to add checkpoints into a stream of data.

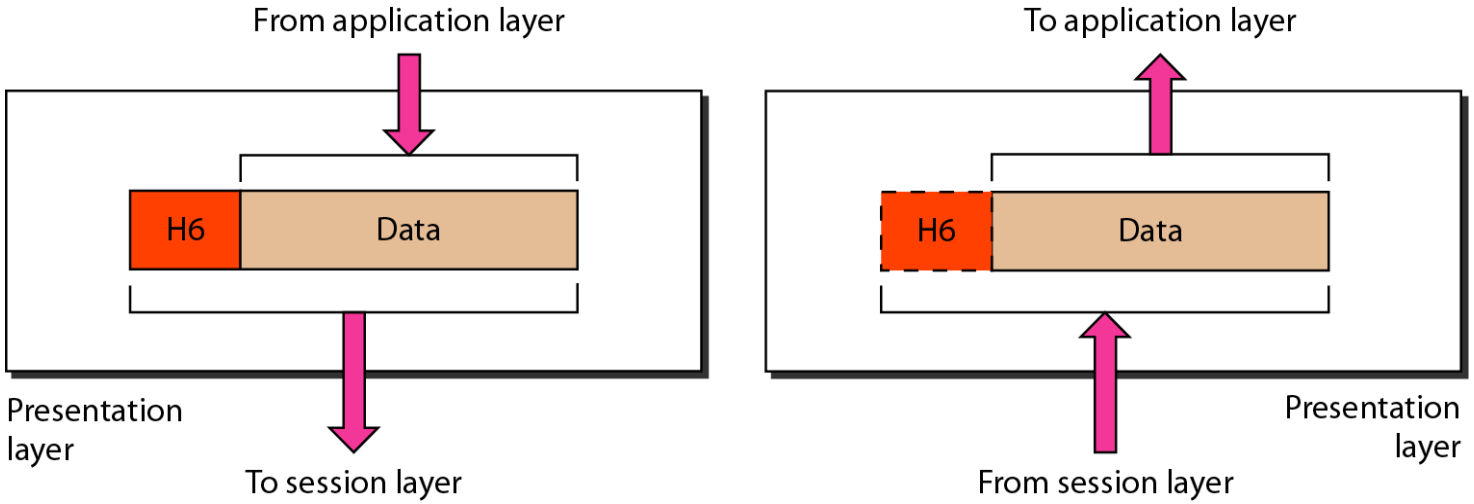


Note

The session layer is responsible for dialog control and synchronization.

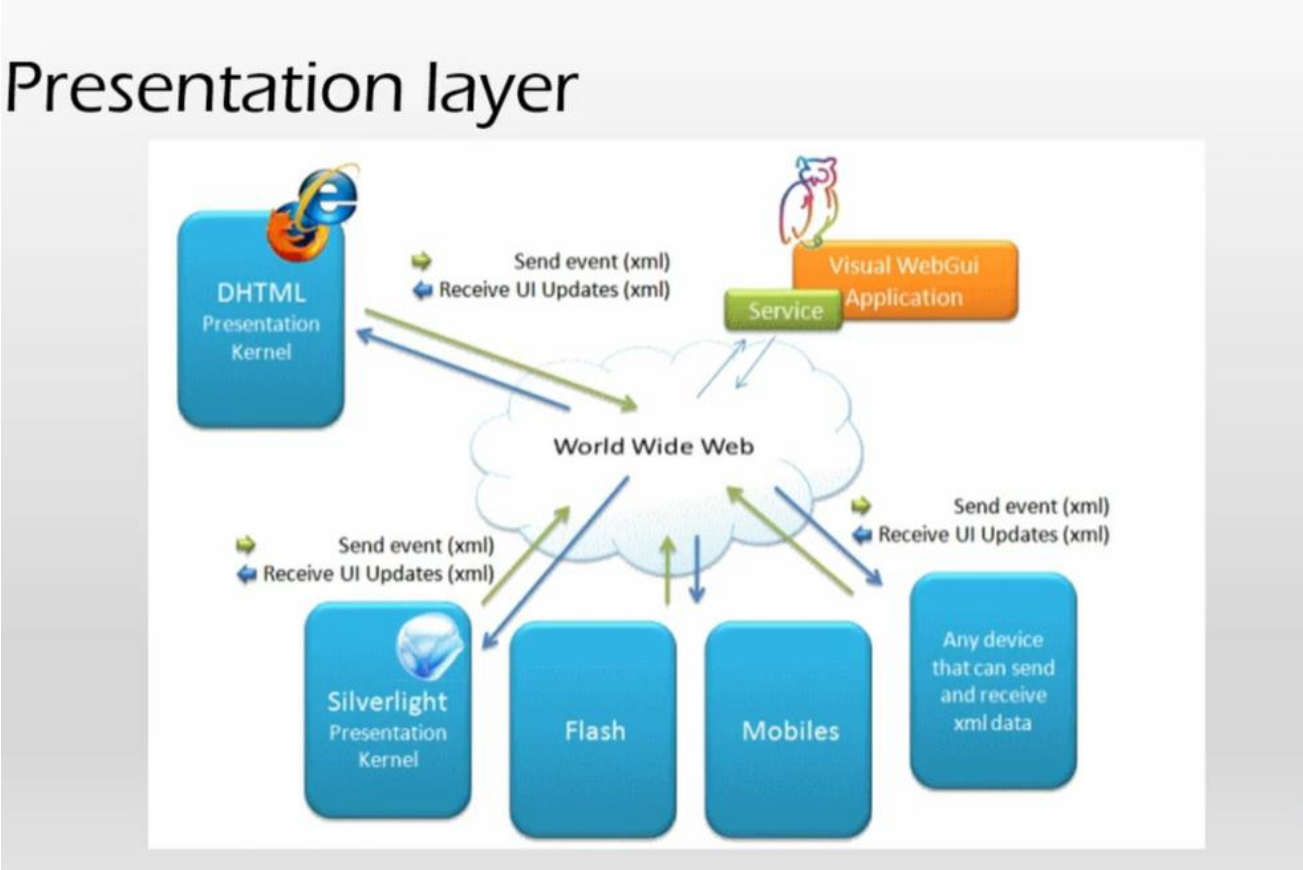
Presentation Layer

- It is concerned with the syntax and semantics of information exchanged between two systems.
- **Translation** – Different computers use different encoding system, this layer is responsible for interoperability between these different encoding methods. It will change the message into some common format.
- **Encryption and decryption**-It means that sender transforms the original information to another form and sends the resulting message over the n/w. and vice versa.
- **Compression and expansion**-Compression reduces the number of bits contained in the information particularly in text, audio and video.



Note

The presentation layer is responsible for translation, compression, and encryption.



Application Layer

- Enables the user, whether human or software to access the network
- Provides user interfaces and support for services such as
  - **FTAM(file transfer, access, management)** - Allows user to access files in a remote host.
  - X.400 **Mail services** - Provides email forwarding and storage.
  - X.500 (**Directory services)** - Provides database sources/distributed information services to access information about various sources and objects.

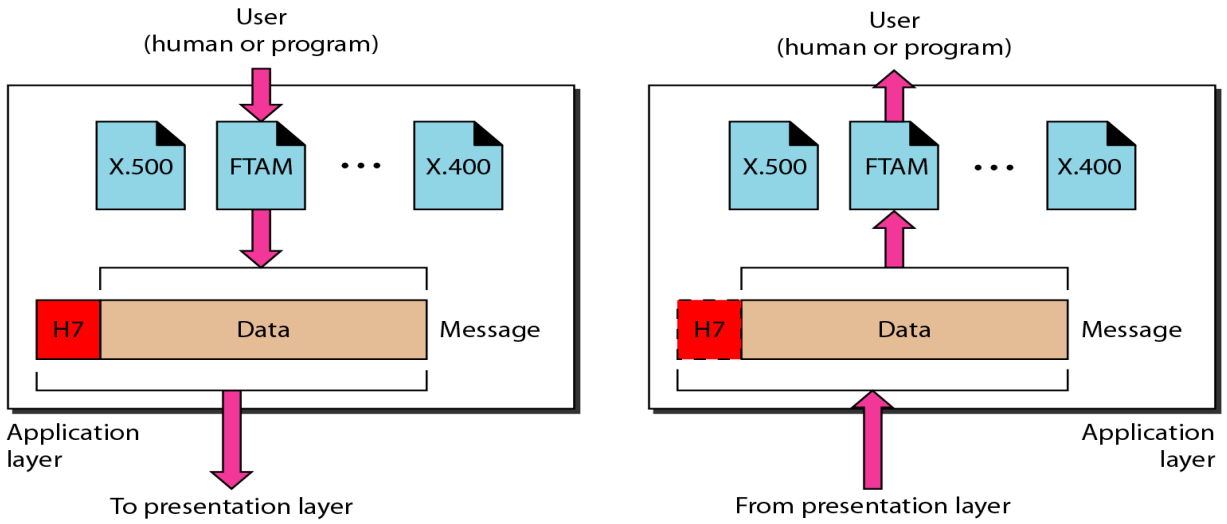
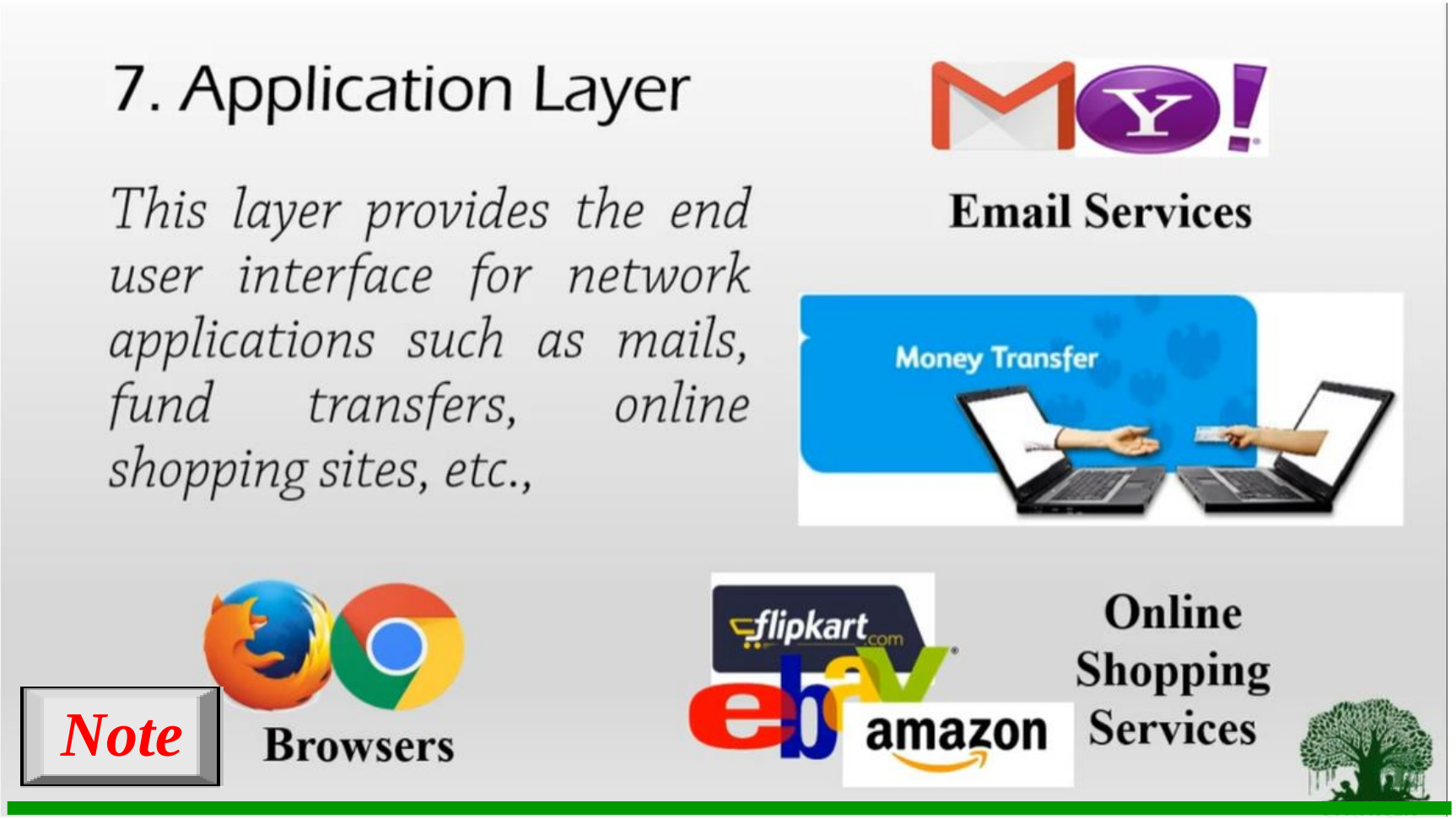
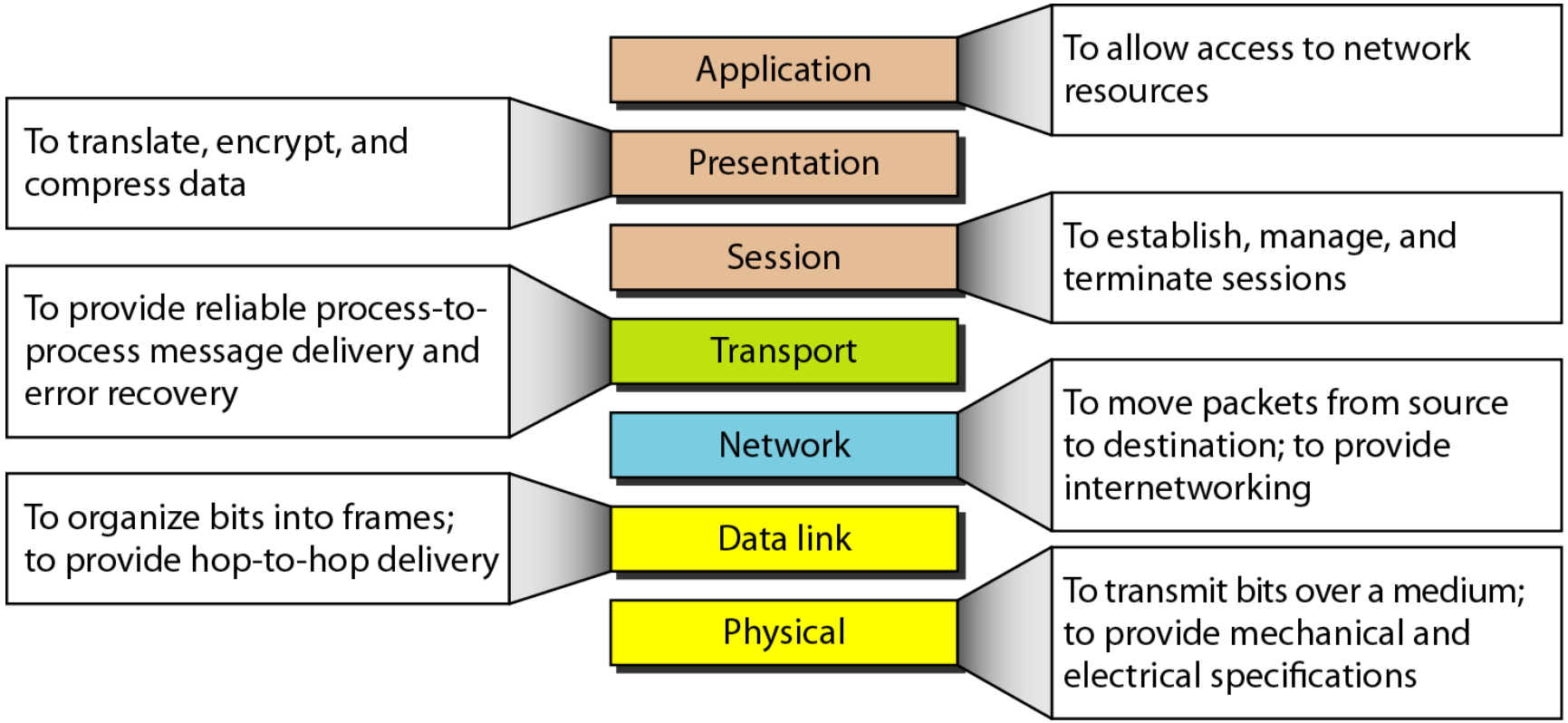


Figure Ex: *Application layer*

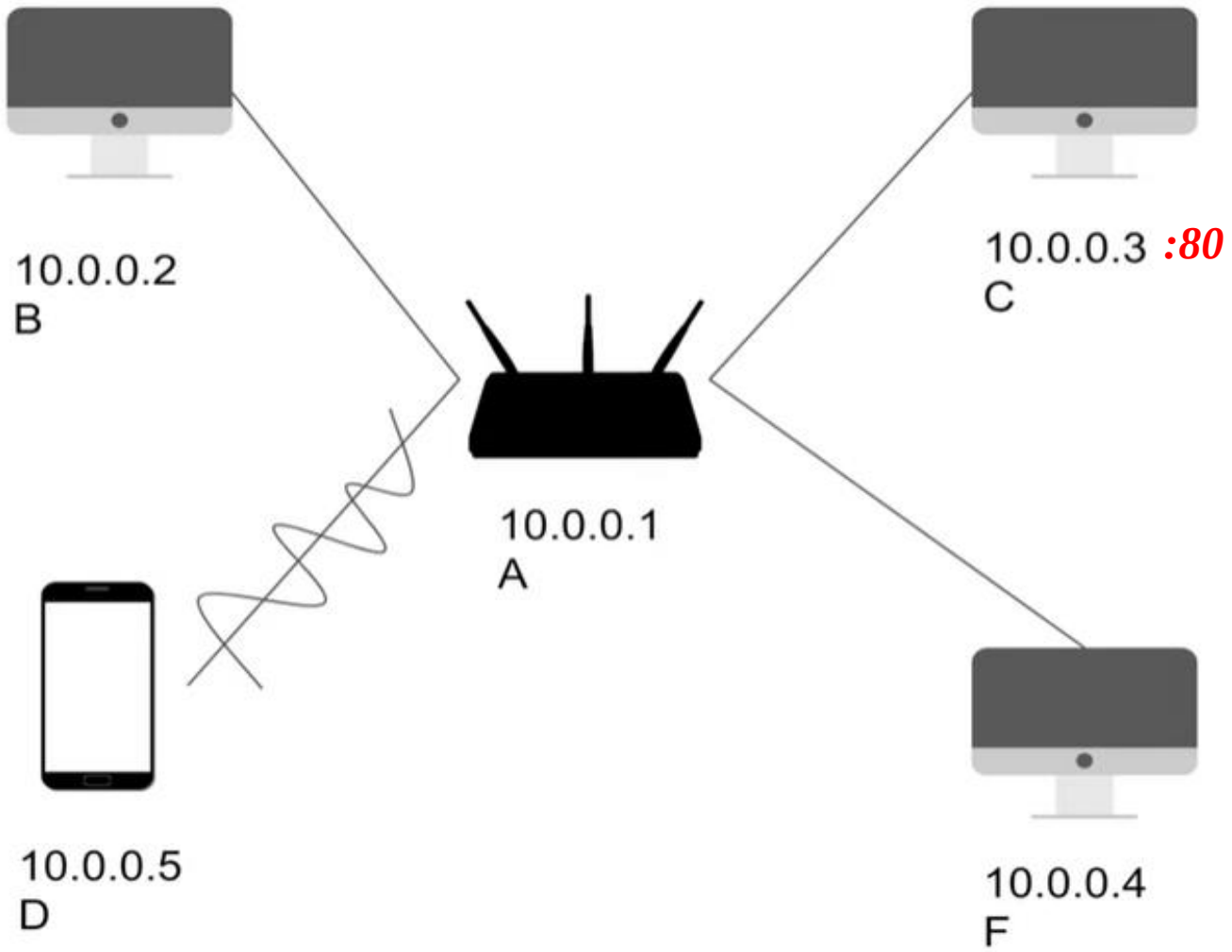


The application layer is responsible for providing services to the user.

Figure *Summary of layers*



10.0.0.3 is a web server having an web app and 10.0.0.5 want to consume it



Layer 7 Application

GET / 10.0.0.3 80
Http headers cookies, content-type etc.

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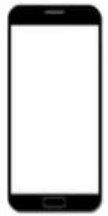
GET / 10.0.0.3 80
Http headers cookies, content-type etc.

Layer 6 Presentation

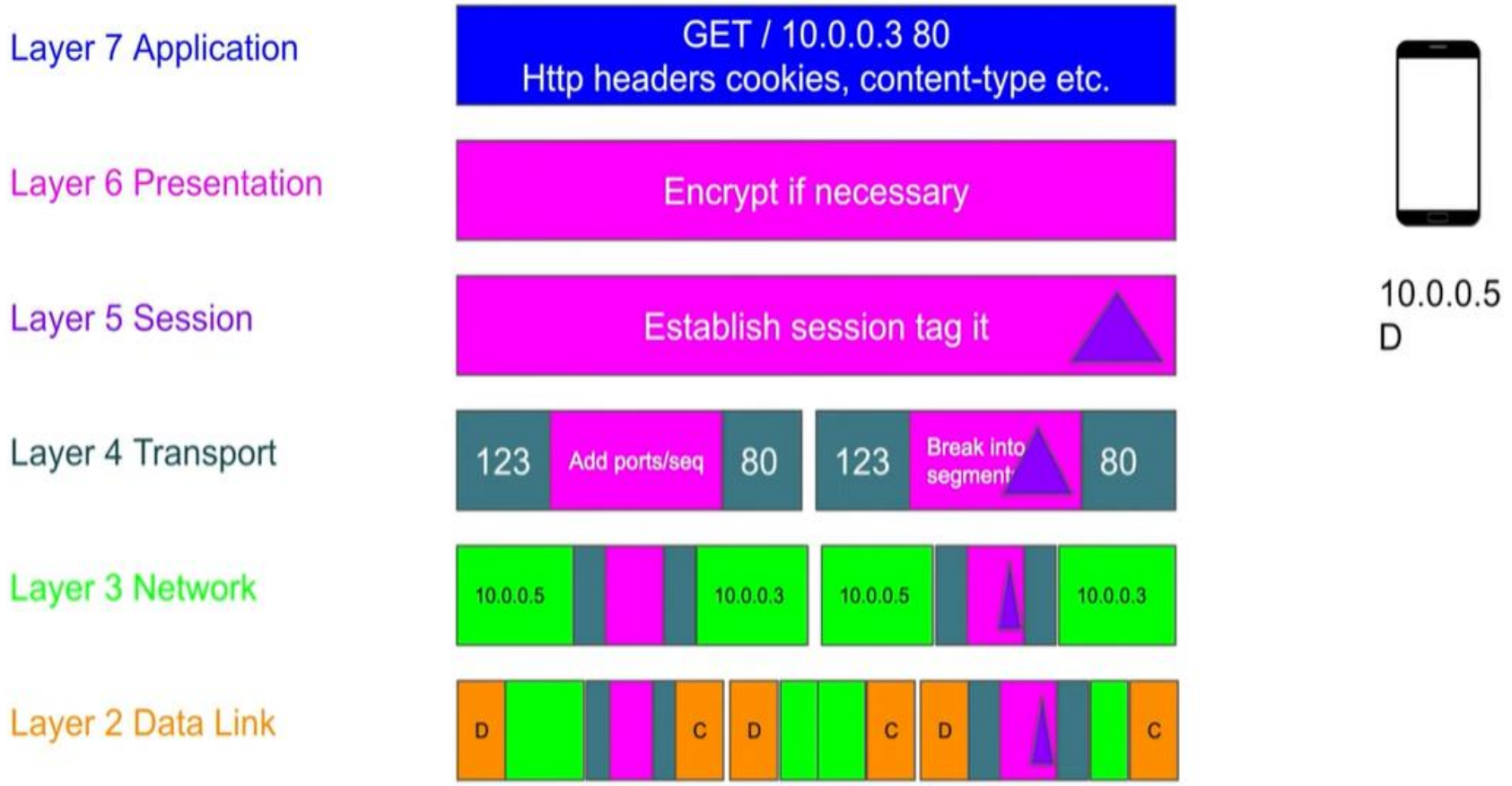
Encrypt if necessary

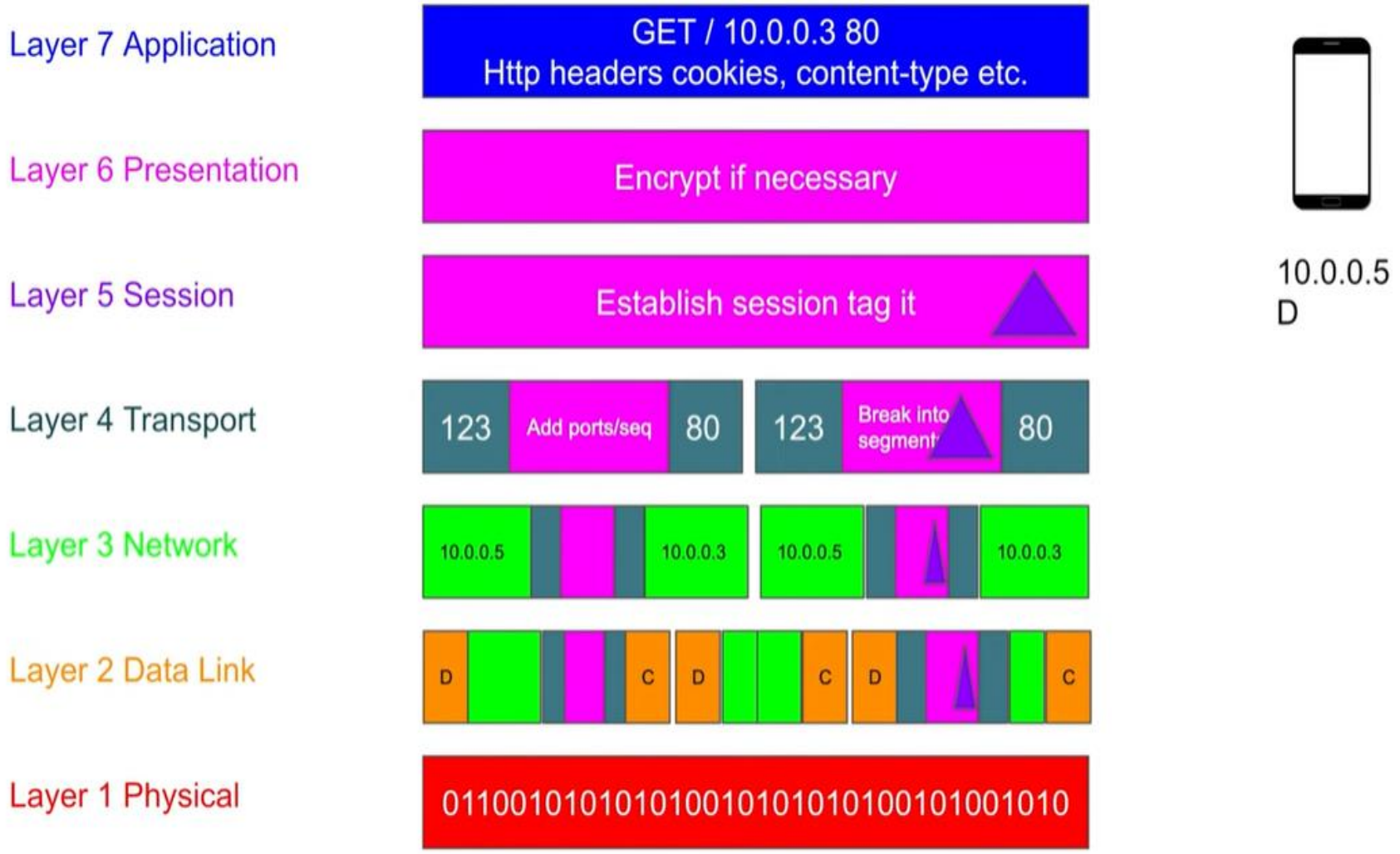


Layer 7 Application
GET / 10.0.0.3 80
Http headers cookies, content-type etc.
Layer 6 Presentation
Encrypt if necessary
Layer 5 Session
Establish session tag it
10.0.0.5
D

Layer 7 Application
GET / 10.0.0.3 80
Http headers cookies, content-type etc.
Layer 6 Presentation
Encrypt if necessary
Layer 5 Session
Establish session tag it
Layer 4 Transport
123
Add ports/seq
80
123
Break into segment
80
10.0.0.5
D

Layer 7 Application
GET / 10.0.0.3 80
Http headers cookies, content-type etc.
Layer 6 Presentation
Encrypt if necessary
Layer 5 Session
Establish session tag it
Layer 4 Transport
123
Add ports/seq
80
123
Break into segment
80
Layer 3 Network
10.0.0.5
10.0.0.3
10.0.0.5
10.0.0.3
10.0.0.5
D


Layer 7 Application
GET / 10.0.0.3 80
Http headers cookies, content-type etc.
Layer 6 Presentation
Encrypt if necessary
Layer 5 Session
Establish session tag it
Layer 4 Transport
123
Add ports/seq
80
123
Break into segment
80
Layer 3 Network
10.0.0.5
10.0.0.3
10.0.0.5
10.0.0.3
Layer 2 Data Link
D
C
D
C
D
C
10.0.0.5
D


Layer 7 Application
GET / 10.0.0.3 80
Http headers cookies, content-type etc.
Layer 6 Presentation
Encrypt if necessary
Layer 5 Session
Establish session tag it
Layer 4 Transport
123
Add ports/seq
80
123
Break into segment
80
Layer 3 Network
10.0.0.5
10.0.0.3
10.0.0.5
10.0.0.3
Layer 2 Data Link
D
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D
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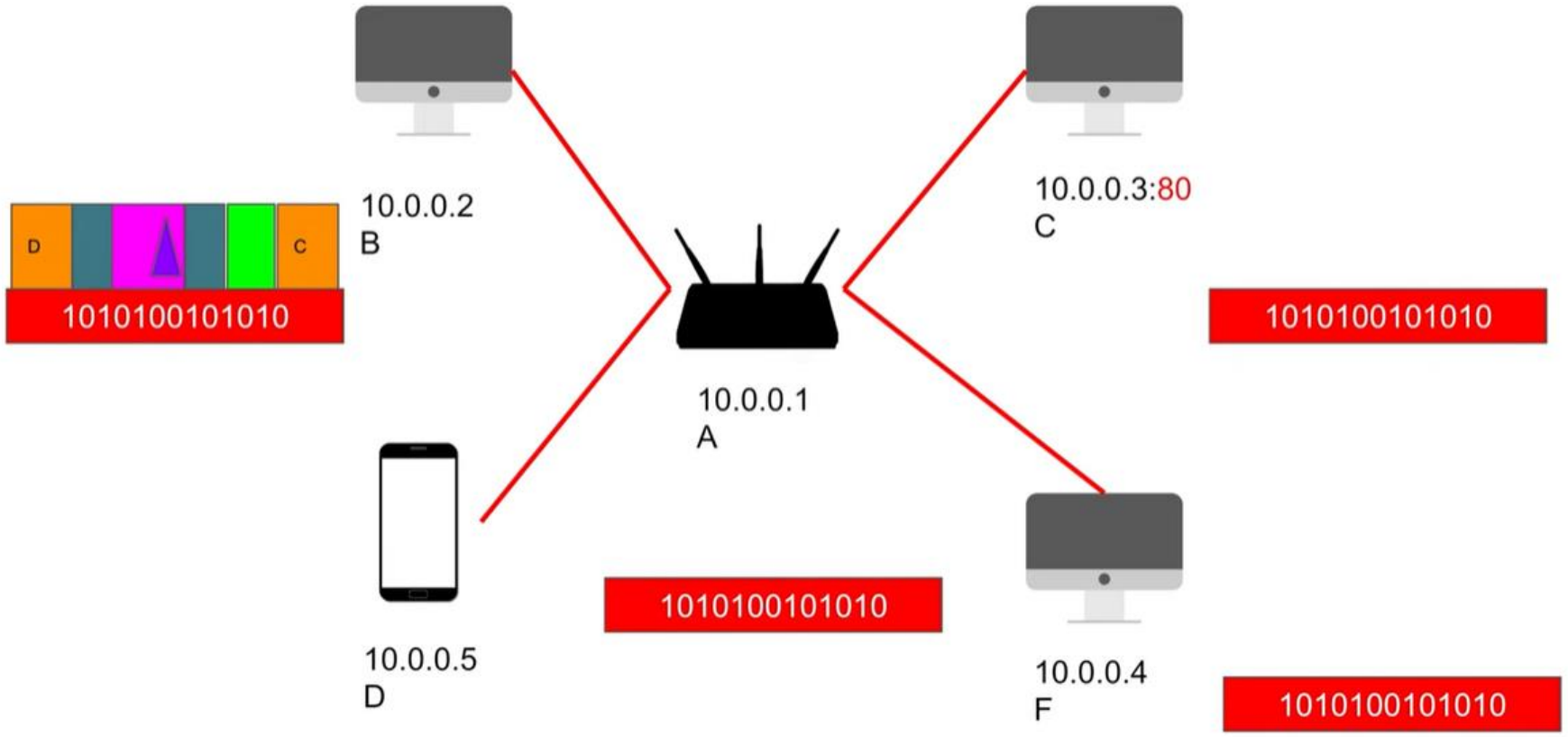
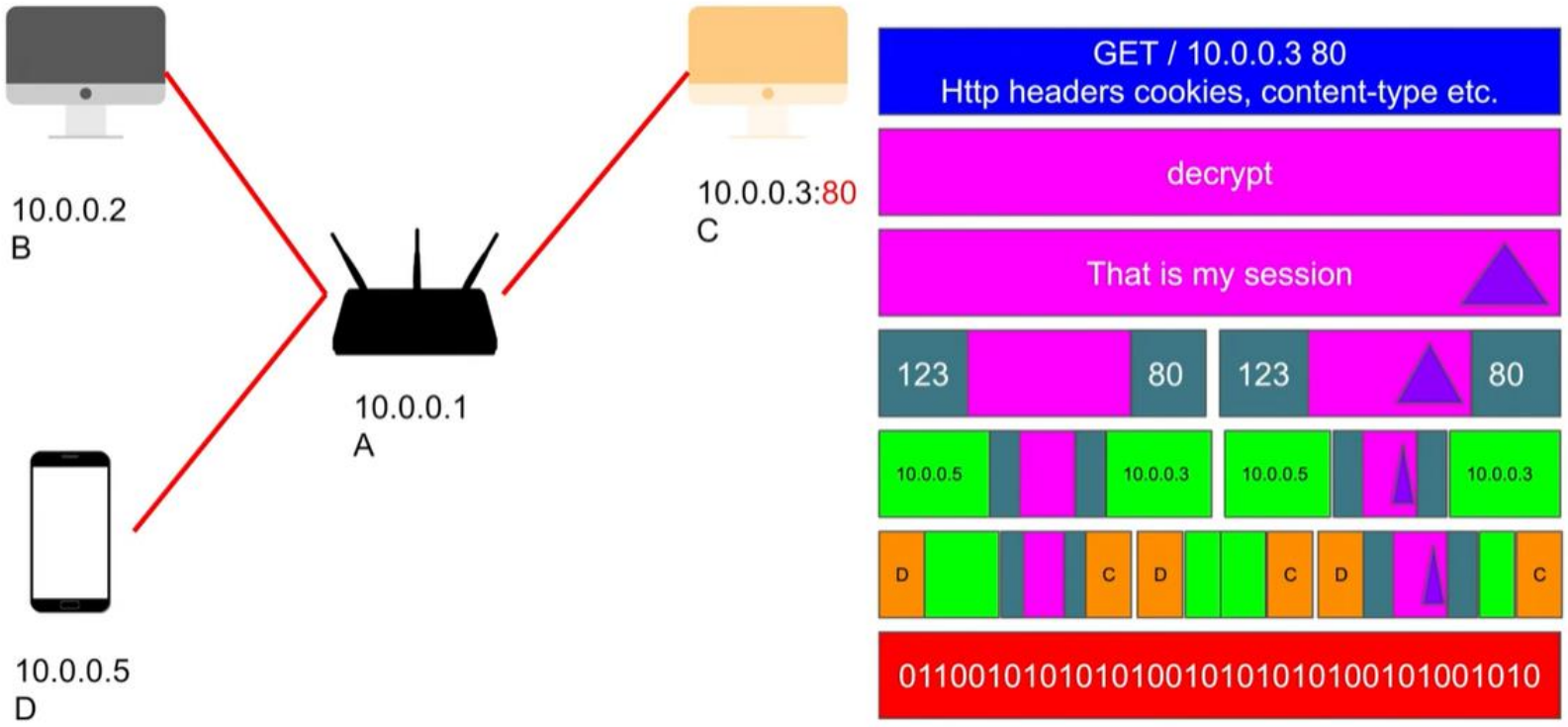
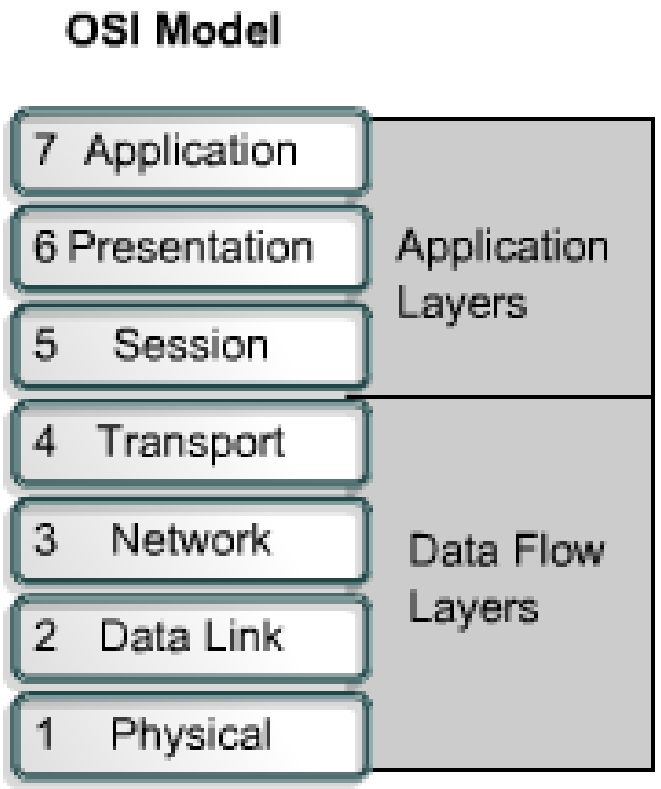
D
C
Layer 1 Physical
0110010101010100101010100101001010
10.0.0.5
D

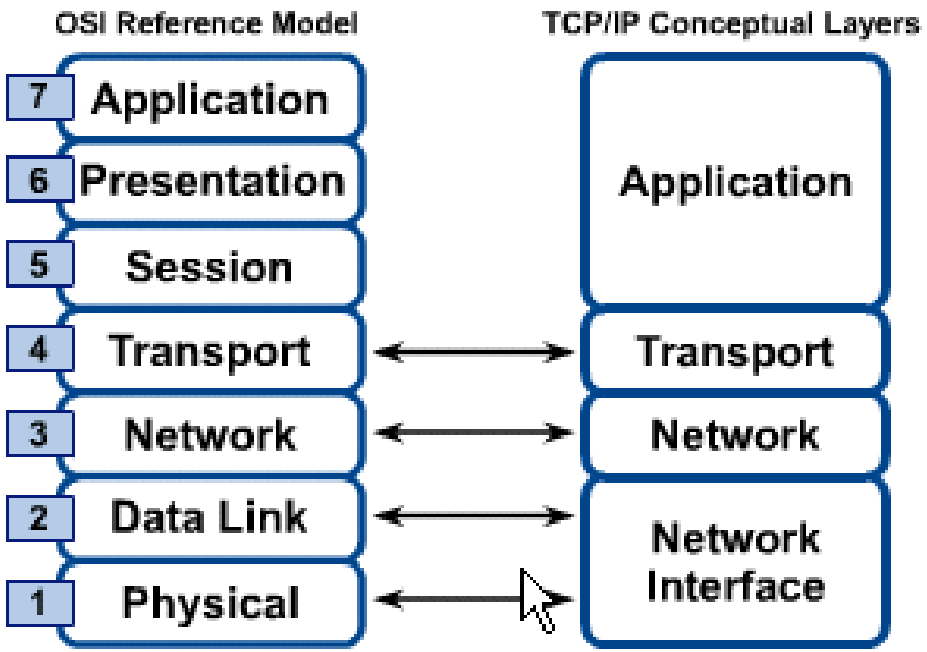

D
C
1010100101010
10.0.0.2
B
10.0.0.3:80
C
1010100101010
10.0.0.1
A
10.0.0.5
D
1010100101010
10.0.0.4
F
1010100101010


10.0.0.2
B
10.0.0.1
A
10.0.0.3:80
C
10.0.0.5
D
GET / 10.0.0.3 80
Http headers cookies, content-type etc.
decrypt
That is my session
123
80
123
80
10.0.0.5
10.0.0.3
10.0.0.5
10.0.0.3
D
C
D
C
D
C
0110010101010100101010100101001010

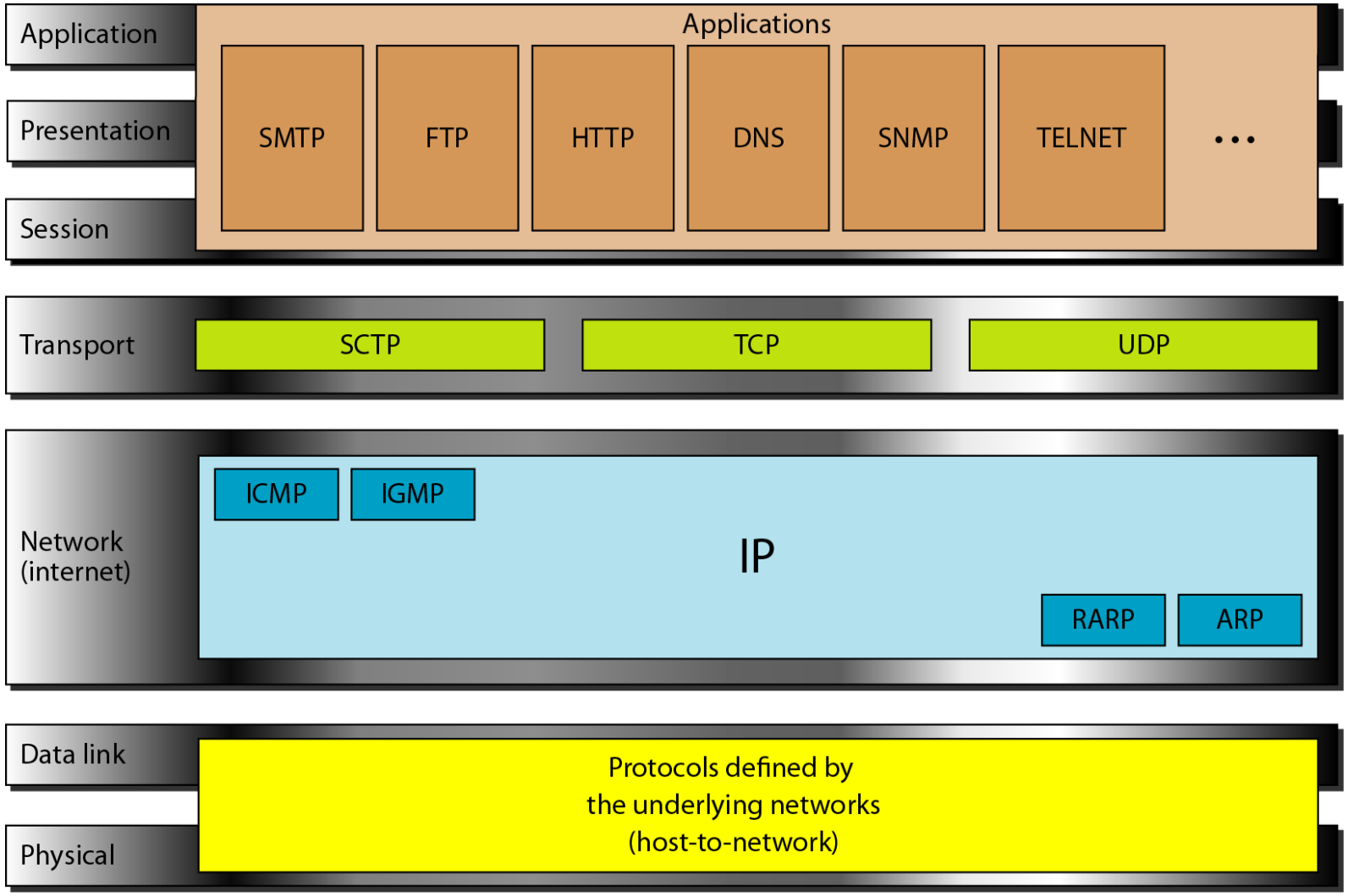
TCP/IP MODEL

OSI & TCP/IP Models

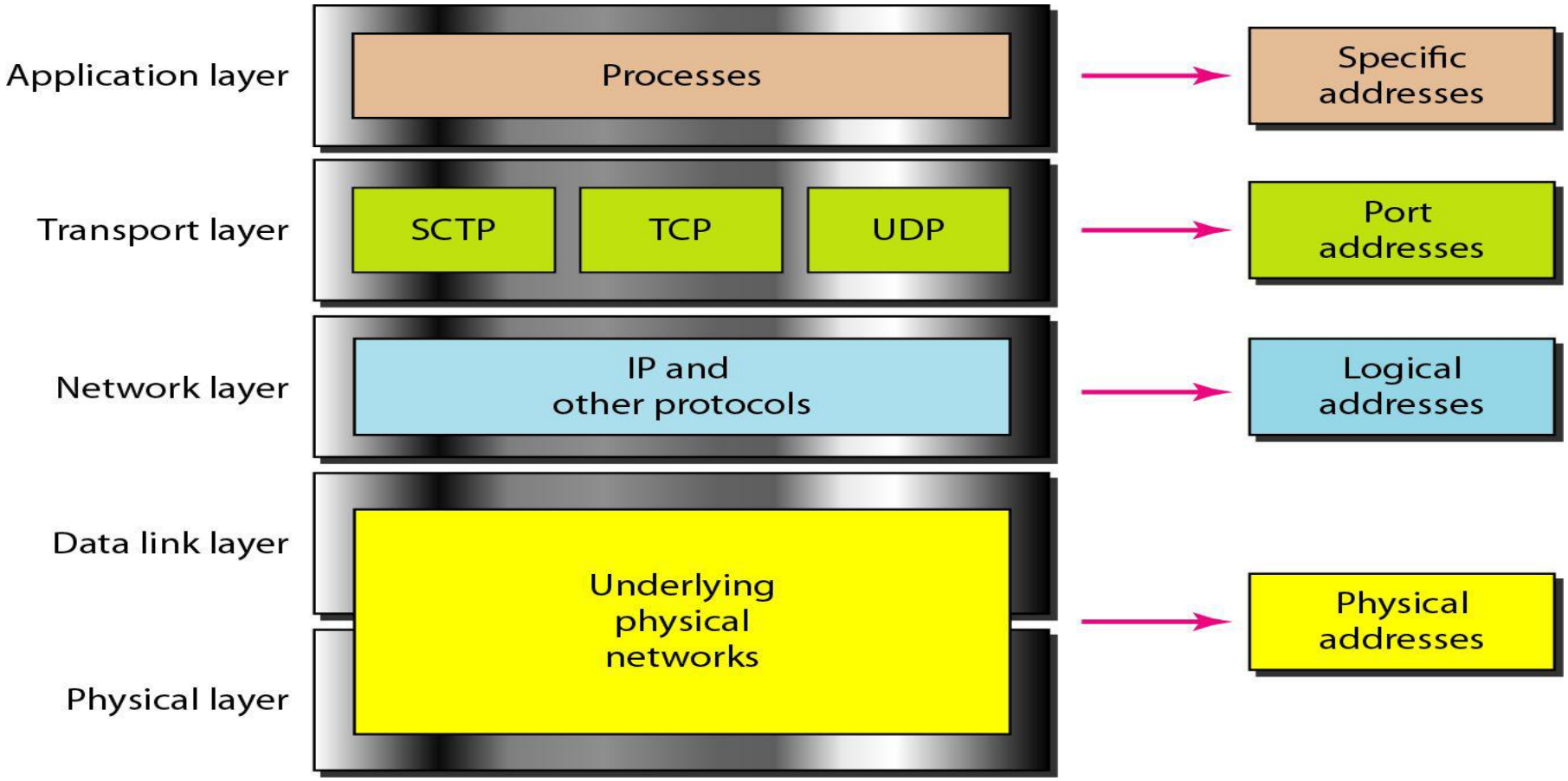




TCP/IP and OSI model



Relationship of addresses to layers



IOT PDS

SDN NFV

Computer Vision

Thank You !!

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